Quantitative Data Analysis – Exercises

(Week 07)

These exercises are about the first three parts of the Data Analysis Process Model (highlighted in yellow in the figure below). The results must be uploaded as separate files (no .zip files) by each student on Moodle. For details on how to submit the results see the tasks below.

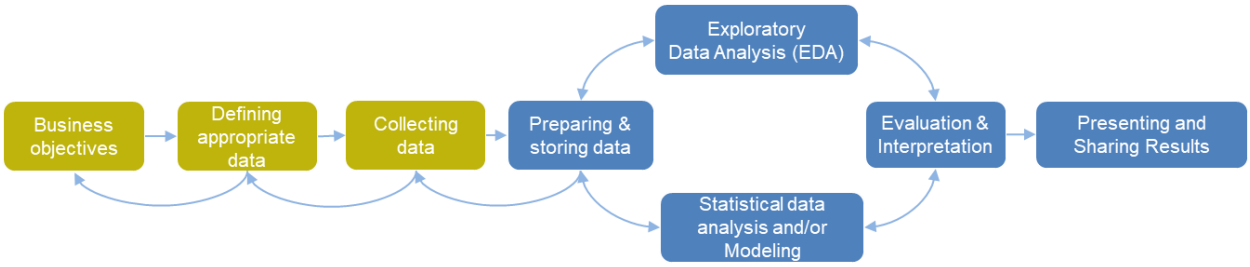


Figure 1: Data analysis process model (see lessons)

Note: It is assumed that basic knowledge of Python and Jupyter notebooks is available from the Statistics module.

Task 1

In this exercise, you must define business objectives and hypotheses for the two use cases defined below. Prepare a short description (bullet points, short notes) by using the following structure:

1. Problem definition (identify one or more potential problem(s)).
2. Objective (define a clear objective based on the problem definition).
3. Research question (formulate a research question based on the objective).
4. Hypothesis (formulate a clear hypothesis based on the research question).
5. Expected added value of the data analysis (business value of data analytics).

The use cases are:

Use-Case (1): Price prediction model for rental apartments in the canton of Zuerich

Use-Case (2): Demand and supply for supermarkets in Swiss municipalities

Expected results: A short description (bullet points) of the problem definition, objective, research question, hypothesis and expected added value for both use cases. The results will be discussed in class.

Task 2

In this exercise, you must propose and describe suitable data for each of the use cases from task 1 and briefly describe the necessary data. You don't have to collect this data in this task; an overview of the data suitable for achieving your business goals is sufficient. Please note that you will need to combine data from multiple sources for the data analysis. For example, it may be helpful to supplement the apartment and supermarket data with municipality-level data. Suitable municipality-level data are the "Regionalporträts 2021: Kennzahlen aller Gemeinden" of the Swiss Federal Statistical Office (SFSO). These data are available as a static file which can be loaded directly with pandas: pd.read_excel('https://raw.githubusercontent.com/mario-gellrich-zhaw/applied_research_methods/master/Week_09/Data/municipality_data_with_taxable_income.xlsx', sheet_name='T21.3.1') for the original SFSO table or sheet_name='data_for_import' for a prepared selection of variables. Data sources for the use cases are:

Use-Case (1): Price prediction model for rental apartments in the canton of Zuerich

https://flatfox.ch

Note: See which information is available per offer.

Municipality-level data: Regionalporträts 2021: Kennzahlen aller Gemeinden (source: SFSO; static file see above)

Use-Case (2): Demand and supply for supermarkets in Swiss municipalities

https://wiki.openstreetmap.org/wiki/DE:Key:shop

Note: See which data is available about supermarkets on OpenStreetMap.

Municipality-level data: Regionalporträts 2021: Kennzahlen aller Gemeinden (source: SFSO; static file see above)

Expected results: An overview (bullet points) of suitable data and data sources for both use cases, including the municipality-level data. The results will be discussed in class.

Task 3

In this exercise, you will practice retrieving data using a Web API and Web Scraping. For use case (2), the data can be obtained via a Web API. Please run the Jupyter notebook 'getting_supermarket_locations_WebAPI.ipynb' for this purpose. The notebook saves the supermarkets available in OpenStreetMap as 'supermarkets.json' in the folder 'Week_07'. Web scraping is necessary for use case (1). In this Web Scraping exercise, we will scrape rental apartment data from https://flatfox.ch for the cities of Winterthur and Zürich. The two cities serve as pilot samples for use case (1), which covers the whole canton of Zürich. The sitemaps (i.e. the configuration of the web scraper) are provided and the selectors are explained in class. Install the 'Web Scraper' extension for Chrome in the Chrome browser. Then perform the following steps:

1. Use Google Chrome to navigate to https://flatfox.ch/de/ and search for rental apartments in Winterthur. Look at the information which is available per offer (e.g. type of object, number of rooms, address and price).

2. Open the Chrome DevTools with CTRL+SHIFT+I (Windows) or CMD+OPTION+I (macOS), switch to the tab 'Web Scraper' and dock the DevTools to the bottom of the browser window (see Figure 2).

3. Import the sitemap for Winterthur: Web Scraper menu -> Create new sitemap (Neue Sitemap erstellen) -> Import sitemap. Paste the content of the file 'flatfox_winterthur_sitemap.json' (folder 'Week_07/Data') and click 'Import sitemap'.

4. Look at the selectors of the sitemap (see Figure 2) and check with 'Data preview' (Datenvorschau) which data are extracted.

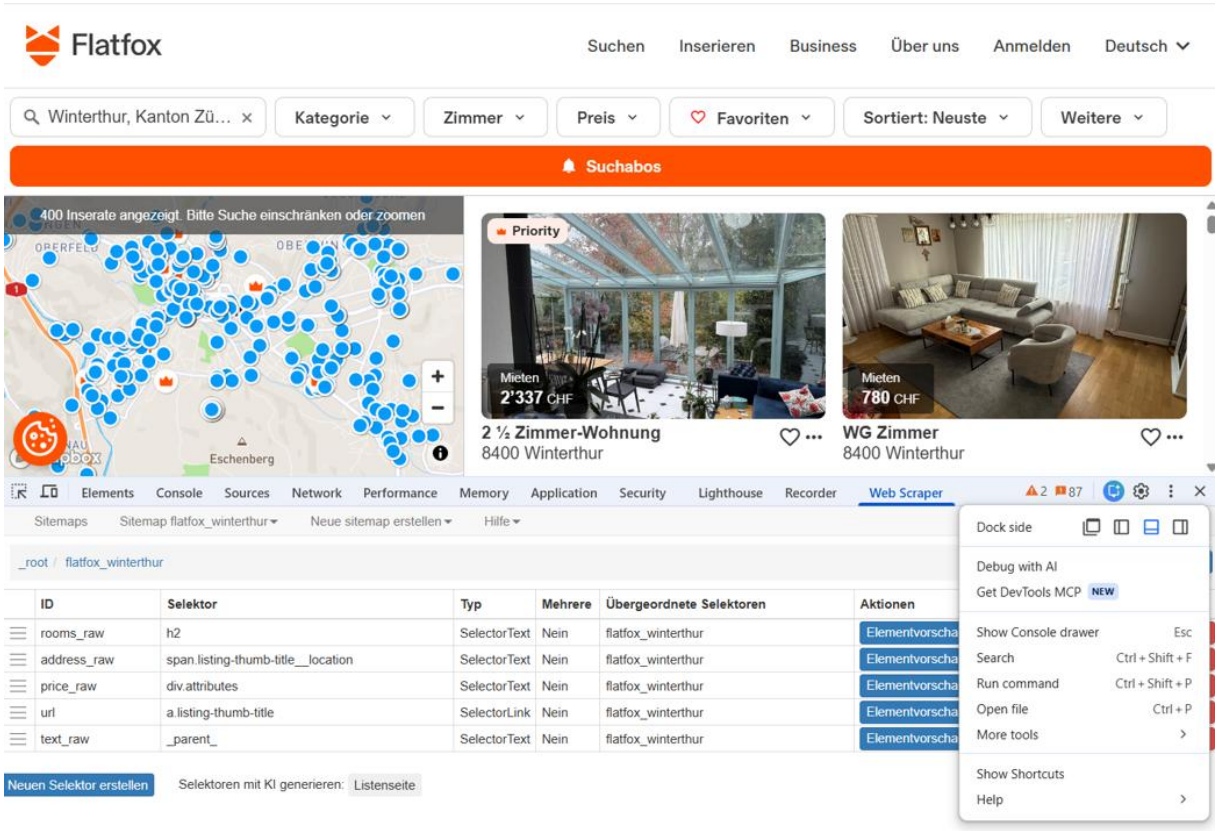


Figure 2: Web Scraper in the Chrome DevTools (docked to the bottom) with the selectors of the Flatfox sitemap.

5. Start the web scraper: Menu -> Sitemap flatfox_winterthur -> Scrape. Set the request interval and the page load delay to 3000 ms and click 'Start scraping'. After scraping, refresh the view using the 'refresh' button.

6. Download the data via Menu -> Sitemap flatfox_winterthur -> Export data as CSV and save it in the folder 'Week_07/Data' as 'flatfox_winterthur.csv'.

7. Repeat steps 3 to 6 with the sitemap 'flatfox_zuerich_sitemap.json' and save the data in the folder 'Week_07/Data' as 'flatfox_zuerich.csv'. Note: The sitemaps scrape the 100 most recent offers per city (one search page without scrolling or pagination), i.e. a sample and not all offers. The search area is a rectangle on the map, so the data also contain offers from neighbouring municipalities as well as WG rooms, parking spaces, commercial spaces and a few offers for sale. These offers must be removed during data preparation. In the following weeks, the datasets provided in the course repository are used.

Expected results:

a) One json file with the supermarket data in Switzerland (supermarkets.json).

b) One csv-file with the scraped rental apartment data of the city of Winterthur (flatfox_winterthur.csv).

c) One csv-file with the scraped rental apartment data of the city of Zürich (flatfox_zuerich.csv).